

Appendix C: Automated Test Suite Inventory

This appendix summarizes the **categories** of tests behind the counts in `??`. Aggregate figures are injected at build time (**804** tests, **47** files; see `src/generate_manuscript_metrics.py` and `output/metrics.json`). Every test uses real mathematical computations—no mocks or fakes.

- ▶ **Categorical axiom tests:** Identity morphism existence, composition associativity, weight invariants, `is_well_formed()` full axiom check
- ▶ **Enriched category tests:** Hom-value constraints, composition inequality, categorical magnitude, magnitude deficit, full composition check, role clustering
- ▶ **Diagram type tests:** Pregroup diagrams validated for `dom == Ty()` and `cod == s`, correct box counts, diagram equality
- ▶ **Metrics tests:** Normal form preservation, depth computation with graceful fallback for pregroup diagrams
- ▶ **Natural transformation tests:** Component morphism construction, `naturality_holds / verify_naturality` on identity and incomplete maps, identity transformation